

London School of Economics
EC400 Mathematics Examination
September 2016

Not for MSc in Finance and Economics Students
Time: 3 hours

This examination has three parts: Revision Mathematics, Mathematics for Microeconomics, and Mathematics for Macroeconomics. Each part is worth 100 marks. Write your name, your class group number, and the name of your class teacher on your script. Explain your answers to get high marks.

Part 1. Revision Mathematics. Answer all questions.

Some questions have yes/no answers, or numerical answers. Explain your answers to these questions.

1. (a) ...

Part 2. Mathematics for Microeconomics. Answer all questions.

5. (25 marks) Consider the following quadratic form

$$Q(\mathbf{x}) = ax_1^2 + bx_2^2 + 2abx_1x_2.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the principal submatrices of A .

- (b) (8 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *positive semidefinite* and *negative semidefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is positive semidefinite or negative semidefinite?
- (c) (9 marks) For what values of the parameter values, a and b , is the quadratic form $Q(\mathbf{x})$ *indefinite*?
- (d) (4 marks) Plot your answer to part (c) in $\mathbb{R}^2$.

6. (25 marks) Answer the following questions:

- (a) (7 marks) Provide two equivalent definitions of *quasi-concavity*.
- (b) (10 marks) Show that any continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, satisfying

$$x \frac{\partial f(x)}{\partial x} \leq 0,$$

must be *quasi-concave*. Prove your answer.

- (c) (8 marks) Find an *increasing* function $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ that is *not quasi-concave*. Prove your answer.

7. (50 marks) For $a > 0$, consider the problem:

$$\begin{array}{ll}\max_{x,y} & ax + y \\ \text{s.t.} & x^2 + (x - y)^2 \leq 1 \\ & x \geq a, y \geq 0.\end{array}$$

- (a) (6 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither.
- (b) (6 marks) Set up the constrained optimization problem using the Lagrangean function. Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of $a > 0$ is Slater condition satisfied?
For parts (c), (d), and (e) suppose that Slater condition is satisfied.
- (c) (8 marks) Using the Kuhn-Tucker approach, write down the necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangean?
- (d) (18 marks) Solve the constrained optimization problem in terms of a .
- (e) (12 marks) State the Envelope Theorem in the context of this problem. Then, find the slope of value function with respect to a without relying on the Envelope Theorem.

Part 3. Mathematics for Macroeconomics. Answer all questions.

8. (50 marks)